Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/14/2011
PI Name:	Collins, Cynthia Ph.D.		
Project Title:	Gravitational effects on quorum sensing and biofilm formation		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	12180-3522	**Congressional District:**	21
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/01/2009	**End Date:**	05/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 5/31/2014 per A. Chu/ARC and NSSC information (Ed., 5/22/13) NOTE: Extended to 5/31/2013 per A. Chu/ARC and NSSC information (Ed., 6/1/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Plawsky, Joel (Rensselaer Polytechnic Institute) Dordick, Jonathan (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNX09AI70G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	There is an urgent need to understand the effects of microgravity on the growth, cellular physiology, and cell-cell interactions in microbial biofilms. This information can then be used to curtail harmful activities of microbial consortia thriving as biofilms on the International Space Station and for the long-term success of human space exploration. Bacterial biofilms were abundant on the MIR space station and were responsible for increasing corrosion and blocking a water purification system. Health and safety hazards linked to the development of biofilms are also of great concern, including increased rates of infection due to the enhanced survival of organisms within biofilms and increased resilience of biofilm populations to antimicrobial compounds. Furthermore, experiments with Escherichia coli have indicated that biofilms formed under microgravity may have even higher levels of resistance to ntimicrobials. We propose to study how gravity alters biofilm formation with the goal of developing new strategies to reduce their impact on the operation of spacecrafts and the health of their crew. We will focus on two model microorganisms that form biofilms both inside and outside of the human body: Pseudomonas aeruginosa and Staphylococcus aureus. These microbes can switch between benign and pathogenic interactions with humans and may be relevant to crew health during extended missions. We will examine how the three-dimensional structures of biofilms formed by these microbes and the requirements and regulation of cell-cell communication systems differ in microgravity compared to normal gravity. We will specifically focus on the role of cell-cell communication in mixed communities where ground-based studies have indicated that inter-species communication can lead to increased virulence. We will employ newly developed, non-leaching, nanotechnology-based, antimicrobial surfaces to aid in assessing differences between biofilm formation and structure in normal and microgravity and to assess their potential to reduce the impact of biolfims in future spacecraft designs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	According to the Center for Disease Control (CDC), hospital-acquired infections are the fourth leading cause of death in the United States behind stroke, cancer and heart disease. Furthermore, it is estimated that >65% of all bacterial infections are associated with biofilms. A greater understanding of biofilms is essential if we are to find effective methods to combat their formation. Furthermore, the low-shear conditions microbes experience in microgravity are similar to those found in the human body that are difficult to study. This work may provide new insights into the role of shear and other physical effects, such as convection, on biofilm formation.
Task Progress:	During the first two years of this project, we have developed new methodologies and have used high-aspect ration vessels to compare biofilm formation in low-shear modeled microgravity. In addition to our ground-based efforts, we had an opportunity to conduct spaceflight experiments on STS-132. These results clearly show that biofilm formation by Pseudomonas aeruginosa is increased during spaceflight, while biofilm formation by Staphylococcus aureus appears to be decreased. Due to the time and effort required to prepare and conduct our spaceflight experiments, we have made somewhat less progress on these ground-based studies than anticipated. However, the flight studies and ground studies together are helping us to elucidate the mechanisms behind the changes in biofilm formation we have observed during spaceflight. The highlights of our ground-based studies are listed below: 1. We have obtained several strains of Staphylococcus aureus and Pseudomonas aeruginosa and shown that both organisms form biofilms in normal and modeled microgravity using high-aspect ratio vessels (HARVs). 2. We have developed a new method for examining biofilm formation on 12 or 13 mm membranes or glass coverslips instead of the proposed glass microcarrier beads . These new surfaces are advantageous because they are easily isolated from the HARVs and they can be easily coated with our antimicrobial surfaces. Mixed cellulose ester membranes and circular glass coverslips have been shown to be the most suitable for our experiments. 3. We have developed procedures for measuring viable cell counts and for confocal imaging of biofilms formed on the membranes or glass slides. Methods for data analysis of our viability studies have been established. The methods for quantitative analysis of confocal images have also been optimized. 4. Methods for RNA isolation and microarray preparation and analysis have been developed. Due to small sample sizes and the large amounts of ribosomal RNA produced by P. aeruginosa, this process has been challenging. However, over the last several months, we have optimized methods for RNA isolation from both model organisms. 5. We have obtained a number of Pseudomonas aeruginosa mutants with impaired motility, including mutants that have no flagella, mutants that make flagella but do not have the motor proteins required for swimming motility, and mutants that are capable of swimming but that do not have the genes necessary for twitching motility that enable movement. These particular mutants will be used to produce a set of data that will used to develop mathematical analyses of the role of motility and physical forces, such as diffusion and convection, on the microbial response to microgravity. 6. Quorum-sensing mutants are also being studies in our HARV system. We have started to work with a lasI deletion mutant that is not able to produce the major acyl-homoserine lactone signal used for density-dependent signaling in P. aeruginosa. 7. In contrast with our spaceflight experiments, we have observed that biofilm formation in vertical and horizontal HARVs (representing modeled microgravity and normal gravity) do not show significant differences in the amount of biofilm biomass observed. We have identified a set of conditions that modulate biofilm formation and virulence gene expression in normal gravity and are beginning to examine if similar effects are observed in modeled microgravity. As such, we are now focusing on changes in phenotype and gene expression. 8. We are actively working to understand the observed differences between biofilm formation in spaceflight compared to modeled microgravity. We hypothesize that mixing and oxygen availability are substantially different between our model system and spaceflight. Both ground-based and spaceflight experiments (Micro-2A aboard STS-135) are underway to elucidate these mechanisms. 9. The process for coating the membranes with nanotechnology-based antimicrobial surfaces has been optimized. At this time, we are focusing our efforts on S. aureus biofilm formation using the lysostaphin-based coatings developed by the Dordick lab. We have switched to using glass slides for these studies, enabling addition controls where the antimicrobial enzyme is attached to the glass surface. We anticipate results from this work in the next few months. 10. At this time, the anti-Pseudomonas coatings are not sufficiently effective in normal gravity to warrant pursuit. Although novel coatings against this organism are of general interest, their design and optimization is beyond the scope of the proposed

	work. 11. We have extended our studies to include a second set of spaceflight experiments that will be flown on STS-135. The Micro-2A experiments will make use of BioServe's flight certified hardware: Group Activation Packs (GAPs) stored in a Commercial Generic Bioprocessing Apparatus (CGBA). The CGBA is a flight certified incubator capable of controlling the temperature between 8°C and 37°C and can hold up to 16 GAPs. Each GAP holds eight Fluid Processing Apparatus (FPA) inserts. The FPA is composed of a glass barrel divided into three chambers that are separated from one another by rubber septa. Each FPA will contain growth medium with membranes in chamber A, a microbial culture suspended in stasis medium in chamber B, and a termination reagent in chamber C. Biofilm formation will be assessed via direct measurements of viable cells and observation of biofilm structures using confocal imaging. RNA will also be isolated for gene expression studies. The Micro-2A experiments will build on key observations made from both our spaceflight and ground-based studies. Several presentations of our work have been made at conferences over the past year (see Bibliography). In addition, we anticipate 4 manuscripts describing this work will be submitted and published in the coming year.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Kim WS, Tengra FK, Young Z, Shong J, Pangule R, Plawsky JL, Dordick JS, Collins CH. "Gravitational effects on biofilm formation by Pseudomonas aeruginosa." Oral Presentation at the 26th Annual Meeting of the American Society for Gravitational and Space Biology (ASGSB), National Harbor, MD, November 4-7, 2010. Program and abstracts. 26th Annual Meeting of the American Society for Gravitational and Space Biology (ASGSB), November 2010. p. 25. , Nov-2010
Abstracts for Journals and Proceedings	Kim WS, Tengra FK, Young Z, Shong J, Pangule R, Plawsky JL, Dordick JS, Collins CH. "Gravitational effects on biofilm formation by Pseudomonas aeruginosa." Poster presentation at the Biofilms IV International Conference, Winchester England, September 1-3, 2010. Biofilms-IV International Conference Proceedings, 2010. p. 89. http://biofilms4.files.wordpress.com/2009/06/biofilms-handbook-final-1-pq.pdf , Sep-2010
Abstracts for Journals and Proceedings	Kim WS, Tengra FK, Young Z, Shong J, Pangule R, Plawsky JL, Dordick JS, Collins CH. "Gravitational effects on biofilm formation by Pseudomonas aeruginosa." Oral Presentation at the 2010 AIChE Annual Meeting, Salt Lake City, Utah, November 7-12, 2010. 2010 AIChE Annual Meeting Conference Proceedings, 2010. http://www3.aiche.org/proceedings/Abstract.aspx?ConfID=Annual-2010&GroupID=1515&SessionID=13969&PaperID=203654 , Nov-2010